



Fig. 2: Uncooled (left) versus cooled (right) camera thermal sensitivity

Tests that can be performed on an IR camera using a blackbody are divided into four categories, as follows:

- Correction and calibration tests: linearity measurement, non-uniformity correction, thermal calibration of signal, dynamic range, etc
- Noise measurement: thermal resolution, noise equivalent temperature difference (NETD), temporal noise, fixed pattern noise, spatial noise, etc
- Spatial resolution and geometrical specifications: MTF, alignment, FOV, magnification, etc
- Range evaluation: minimum resolvable temperature difference (MRTD), detection, recognition and identification (DRI) ranges, etc

Performance measurement parameters

Several factors influence the performance ranges, including spatial resolution, thermal sensitivity of the IR camera, FOV, shape and camouflage of the target, spectral band of detection and experience of the operator. To compare the performance of a system from different manufacturers, establishing a common test frame is required, in addition to listing all conditions influencing DRI range values.

NATO has developed STANAG 4347 standard that defines the target parameters resulting from size, shape, temperature, material properties and emissivity. It also defines conditions of use resulting from ambient temperature and type of background scene to take into account in MRTD computation, giving values of DRI ranges. This standard has been widely adopted in the thermal imaging industry.

MRTD is a standard performance measurement for thermal imagers. This measurement leads to determination of DRI ranges of the IR camera

under test. DRI ranges—expressed in kilometres—can be found in the specification table of IR camera brochures. To select the right sensor that meets application requirements, it is essential to first define DRI ranges perfectly and then assess with regards to globally-adopted industrial standards.

Detection is the ability to distinguish an object from the background; for example, to detect a target several kilometres out of the background.

Recognition is the ability to classify the object class (animal, human, vehicle, boat, etc); for example, the target is recognised as a human walking along the fence.

Identification is the ability to describe the object in detail (man with hat, deer, jeep, etc); for example, two males wearing trousers and jackets are identified, and one is smoking.

John Johnson, a scientist from Army Night Vision Laboratory, has defined thresholds as the minimum number of line pairs to detect, recognise or identify targets captured by scene images. According to Johnson's criteria, DRI lower limits are typically in the following range:

- Detection limit: 0.75/1lp
- Recognition limit: 3/4lp
- Identification limit: 6/8lp

where lp is the line pair as one white line adjacent to a black line

MRTD measurement is performed using a differential blackbody as the IR reference source, a four-bar target as the reference object, positioned in front of the emissive surface of the reference source and a collimator projecting the object in front of the camera under test. A differential blackbody enables to set a positive or negative temperature difference between target bars and their background. The new generation of electronic controllers at the heart of the

blackbody control and monitoring are ideally suited for testing and calibration of IR systems.

NETD can be considered as the thermal resolution that measures the temperature difference between an object and its environment, required to generate a variation of the IR camera signal equal to its temporal noise. The camera's ability to capture tiny details at great distances is defined as spatial resolution. It is closely related to the number of camera pixels—the more the pixels, the better the spatial resolution will be and the greater the detection range.

Thermal sensitivity is the ability to distinguish tiny differences in temperature. It has a high impact on detection performance. Cooled IR cameras provide better thermal sensitivity than uncooled ones for the same number of pixels.

Effects of atmosphere

To sense the temperature remotely, you must account for a variety of surface and environmental conditions. Earth's atmosphere will both absorb and emit IR radiation based on its ambient temperature, air density, smoke, snow, humidity, rain, fog, dust, and the distance between the camera and surface.

The amount of absorption or emission of IR radiation from the atmosphere has a marked effect on thermal imaging readings, so that must be taken into account. If measurements at longer distances are unavoidable, atmospheric conditions must be characterised and taken into account to calculate atmospheric transmission.

In addition, an increased number of pixels are required where thermal images are out of focus due to target or imager motion to make accurate radiometric measurements. There are various factors that can affect the accuracy of radiometric surface temperature measurements. Surfaces with high reflection and low emissivity can reduce the impact of reflections, exacerbated oblique reflections, impact of sun glints and background temperature reflection. This could also be achieved by combining a rough surface texture with high emissivity to cancel out the impact of high reflectivity and uncertain emissivity. **EFY**